A Project Report on

AI DRIVEN FACIAL EXPRESSIVE STATE AUDIO CURATION SYSTEM
Mood To Music

A Dissertation submitted to JNTU Hyderabad in partial fulfillment of the academic requirements for the award of the degree.

Bachelor of Technology

In

Computer Science and Engineering

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CERTIFICATE

This is to certify that the Major Project report entitled "AI DRIVEN FACIAL EXPRESSIVE STATE AUDIO CURATION SYSTEM" being submitted by M. Sai Prasanna (23H55A0520), M. Sheshank Reddy (23H55A0521), O. Sravan Kumar (23H55A0523), P. Sai Pavan (23H55A0524) in partial fulfillment for the award of Bachelor of Technology in Computer Science and Engineering is a record of bonafide work carried out under my guidance and supervision.

The results embodies in this project report have not been submitted to any other University or Institute for the award of any Degree.

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ABSTRACT

This project presents an intelligent, real-time AI-driven system that automatically curates audio content based on a user's facial emotional state. The proposed framework integrates computer vision, deep learning, and affective computing to analyse facial expressions captured through a live webcam feed. A hybrid model combining Haar-Cascade face detection with Convolutional Neural Networks (CNN) or deep feature extractors such as ResNet is employed to classify emotions including happiness, sadness, anger, surprise, and neutrality. Once the emotional state is detected, an audio curation module dynamically maps the predicted emotion to an appropriate music category, enabling mood-aware and personalized audio recommendations.

The system aims to reduce manual searching effort and enhance user experience by providing instant, emotion-adaptive music suggestions. Designed for efficiency and real-time performance, the model delivers fast predictions with high accuracy, making it suitable for entertainment platforms, wellness applications, and personalized media systems. This work demonstrates the potential of combining facial affect recognition with multimedia recommendation engines to create intuitive, user-centric emotional computing applications.

CHAPTER 1
INTRODUCTION

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1.1. Problem Statement

With the rapid growth of digital music platforms, users often struggle to choose audio content that matches their real-time emotional state, as traditional recommendation systems rely mainly on past preferences or manual inputs. Since facial expressions naturally convey mood, an AI-based system capable of detecting emotions from facial cues can greatly improve personalization. However, variations in lighting, subtle expressions, and real-time processing make accurate emotion-based audio curation challenging, creating the need for an intelligent system that can reliably analyze facial expressions and automatically recommend suitable music.

1.2. Research Objective

The objective of this project is to develop an AI-driven system that interprets a user's facial expressions in real time and curates appropriate audio content based on the detected emotional state. The system integrates computer vision for face detection with deep learning models for emotion classification to identify moods such as happiness, sadness, anger, and neutrality. It also includes an emotion-to-audio mapping module that selects music corresponding to the user's mood, aiming to deliver a fast, accurate, and personalized audio experience.

1.3. Project Scope and Limitations

Project Scope

The scope of this project includes developing an AI-based system capable of detecting a user's facial emotions in real time and curating suitable audio content based on the identified emotional state. The system integrates face detection, deep learning–based emotion classification, and an audio recommendation module to provide personalized, mood-aware music suggestions. It focuses on major emotion categories such as happy, sad, angry, neutral, and surprise, and operates through a simple user interface for easy interaction. The project aims to enhance user experience by offering automatic, emotion-driven audio curation without requiring manual input or prior listening history.

Limitations

The system's accuracy depends heavily on camera quality, lighting conditions, and clear visibility of the user's face. It may misclassify subtle, mixed, or partially hidden expressions, and performance can vary across different skin tones, ages, or facial features without a sufficiently diverse training dataset. The model may also struggle with rapid emotional changes or occlusions such as masks and glasses. Additionally, real-time processing requires moderate computational resources, limiting deployment on low-power devices. The system currently focuses only on facial expressions and does not incorporate additional emotional cues such as voice, behavior, or physiological signals.

CHAPTER 2
BACKGROUND WORK

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BACKGROUND WORK

2.1 Music Recommendation System Based on Emotion, Age & Ethnicity

2.1.1 Introduction

This system detects a user's emotion, age, and ethnicity from a webcam image using three CNN-based models. Emotion is classified using FER-2013, ethnicity using the AGE dataset, and age using regression. The results are integrated with a web application that uses Spotify APIs to recommend music based on demographic and emotional states.

2.1.2 Merits, Demerits and Challenges

Merits

• Combines three models for better personalization.

• Provides accurate emotion and ethnicity detection.

• Works in real-time using webcam input.

• Easy to deploy through Flask + ASP.NET MVC.

Demerits

• Requires good lighting and clear facial images.

• Age prediction has higher error due to dataset limitations.

• Models trained on grayscale images → no transfer learning.

• Performance depends on camera quality.

Challenges

• Improving accuracy in low-light environments.

• Expanding datasets for better generalization.

• Ensuring real-time speed without lag.

• Handling variation in facial angles and expressions.

2.1.3 Implementation

• CNN models trained using FER-2013 & AGE datasets.

• Flask API returns emotion, age & ethnicity predictions.

• ASP.NET MVC UI shows results & fetches playlists via Spotify API.

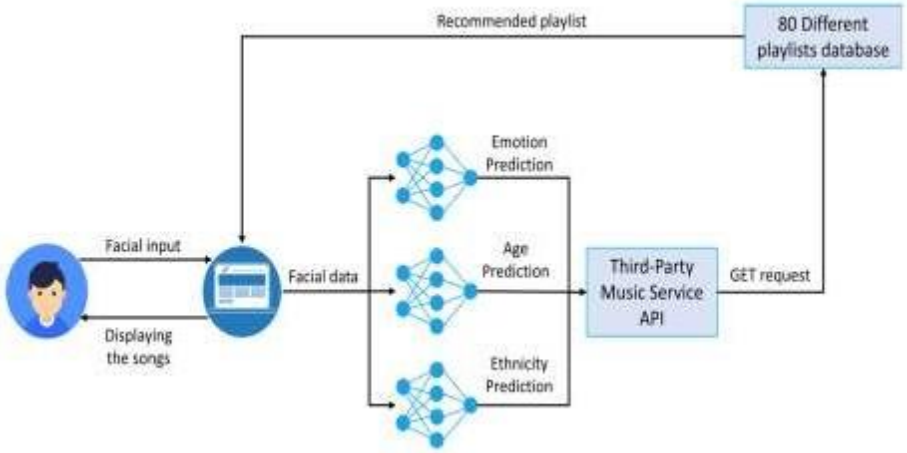


Fig 2.1 Block Diagram of Existing System

2.2 Emotional Detection & Music Recommendation System Based on Facial Expression

2.2.1 Introduction

This method uses facial expression recognition to detect user emotions such as happy, sad, or angry. Face detection and landmark extraction, the system predicts emotion using models trained on CK+ and HELEN datasets. Based on the detected emotion, a suitable playlist is displayed through a web application.

2.2.2 Merits, Demerits and Challenges

Merits

• Real-time emotion detection through webcam.

• Uses facial landmarks for higher accuracy.

• Easy-to-use music recommendation interface.

• Works well for basic emotion categories.

Demerits

• Highly dependent on good lighting.

• Limited to a small set of emotions.

• Subtle expressions are harder to classify.

• Small dataset limits generalization.

Challenges

• Increasing accuracy in low-light and noisy backgrounds.

• Handling fake or forced expressions.

• Improving dataset diversity.

• Ensuring stable predictions for moving faces.

2.2.3 Implementation

• Landmark extraction from HELEN dataset.

• Emotion classification using CK+ dataset.

• Frontend built with PHP, MySQL, HTML, CSS.

• Emotion-based music playlist displayed instantly to user.

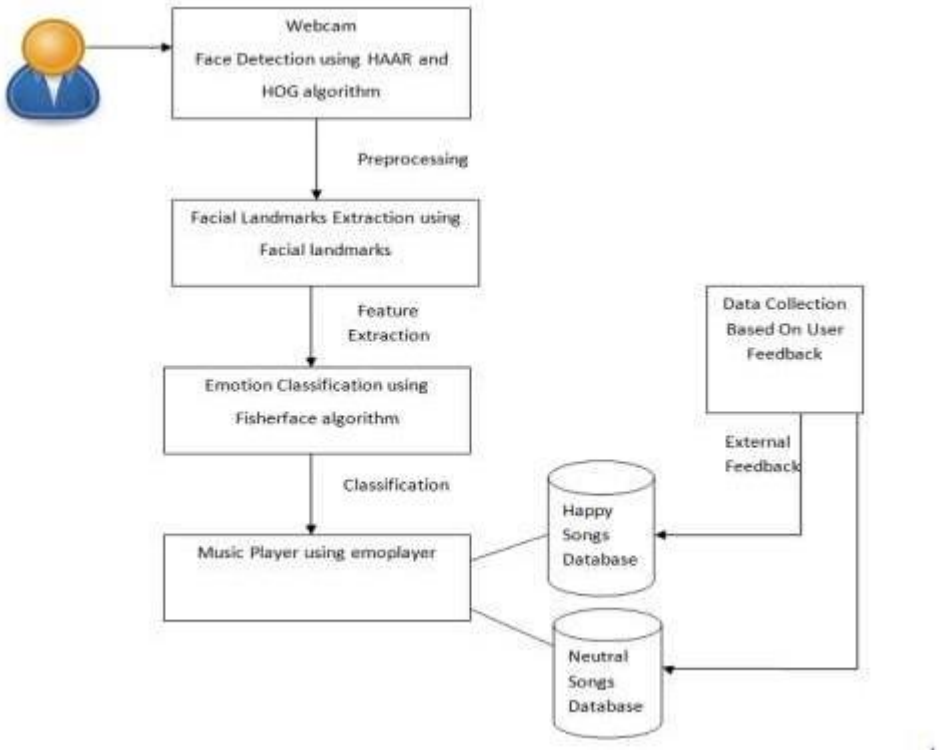


Fig 2.2 Architecture of Music Recommendation System Using Face Expressions

2.3 Existing Method 3: Music Emotion Prediction Using Recurrent Neural Networks (RNNs)

2.3.1 Introduction

This method uses Recurrent Neural Networks (RNNs) and their advanced variants—Bidirectional RNNs (BRNNs) and Long Short-Term Memory networks (LSTMs)—to classify emotions conveyed in music for recommendation systems. The approach analyzes the temporal patterns in audio signals by extracting spectral, rhythmic, and expressive audio features using libraries such as Librosa. The system then maps each audio clip into emotional quadrants based on **Russell's Emotion Model**, enabling emotion-aware recommendation. The paper evaluates different RNN architectures on datasets of 900 audio clips, augmented datasets (3600 clips), and larger datasets (14,000+ tracks), demonstrating how temporal modeling improves emotion classification in music

2.3.2 Merits, Demerits and Challenges

Merits

• Captures temporal dependencies in music signals better than static models.

• LSTM and BRNN architectures outperform standard RNNs on larger datasets.

• Works effectively with extracted spectral features (MFCC, chroma, mel-spectrogram, tonnetz).

• Data augmentation significantly improves accuracy by 20–30%.

• Suitable for music emotion tagging and emotion-aware recommendation systems.

Demerits

• Standard RNNs perform poorly on larger datasets (accuracy drops with size).

• Training is computationally expensive and time-consuming due to sequential processing.

• Highly sensitive to dataset imbalance and quality of audio feature extraction.

• Deep models like LSTM need more data and tuning compared to traditional classifiers.

• May be outperformed by simpler machine learning models on small datasets due to overfitting risks.

Challenges

• Identifying optimal sequence length and feature representation for each audio clip.

• Preventing gradient vanishing/explosion during RNN training.

• Mapping music genres or moods into the correct emotional quadrants.

• Managing resource limitations when training on 14,000+ audio tracks.

• Achieving stable accuracy across different music styles, genres, and variable clip durations.

2.3.3 Implementation of RNN-Based Music Emotion Recognition

• Audio clips (30 seconds each) are collected and labeled according to Russell's four emotional quadrants.

• Features are stacked into 3D sequences with shapes like *(900, 204, 1295)* or *(3600, 204, 1290)* for feeding into RNNs

• Three models are trained:

– Standard RNN

– Bidirectional RNN (BRNN)

– Long Short-Term Memory (LSTM)

• AdamW optimizer and cross-entropy loss are used; gradient clipping is applied for stability.

• A 5-fold cross-validation process evaluates model robustness across datasets.

• Results show BRNNs and LSTMs outperform vanilla RNNs, especially on large and augmented datasets, achieving up to 65% accuracy on four emotion classes

CHAPTER 3
PROPOSED SYSTEM

CHAPTER 3
PROPOSED SYSTEM

3.1 Objective of Proposed Model

The objective of the proposed system is to develop an intelligent, real-time AI model that detects a user's emotional state through facial expressions and automatically curates suitable audio content based on the detected mood. The system aims to integrate deep learning–based facial emotion recognition with a dynamic audio-mapping engine to deliver highly personalized, mood-aware music suggestions. The goal is to ensure fast processing, reliable predictions, and an enhanced user experience by minimizing manual input and providing emotion-driven audio recommendations.

3.2 Algorithms Used for Proposed Model

1. Haar-Cascade Face Detection

Used for detecting the user's face region from live webcam input. It enables fast and efficient face localization with minimal computational cost.

2. Convolutional Neural Networks (CNN)

CNNs are used to extract spatial facial features and classify emotions such as happy, sad, angry, neutral, and surprise. This forms the core emotion recognition model.

3. ResNet / MobileNet (Deep Feature Extractor)

A lightweight deep learning model used for improving accuracy, robustness, and performance across different lighting conditions and face variations.

4. Softmax Classification Algorithm

Converts the model output into probability scores for each emotion class, selecting the emotion with the highest probability.

5. Emotion–Audio Mapping Algorithm

Maps each detected emotion to a predefined music category, such as:

- Happy → Energetic Playlist
- Sad → Calm/Soothing Playlist
- Angry → Relaxation Playlist
- Neutral → General Playlist
- Surprise → Upbeat Playlist

6. Optimization Algorithms (Adam / SGD)

Used during training to minimize loss and improve model convergence, ensuring high prediction accuracy.

3.3 Designing

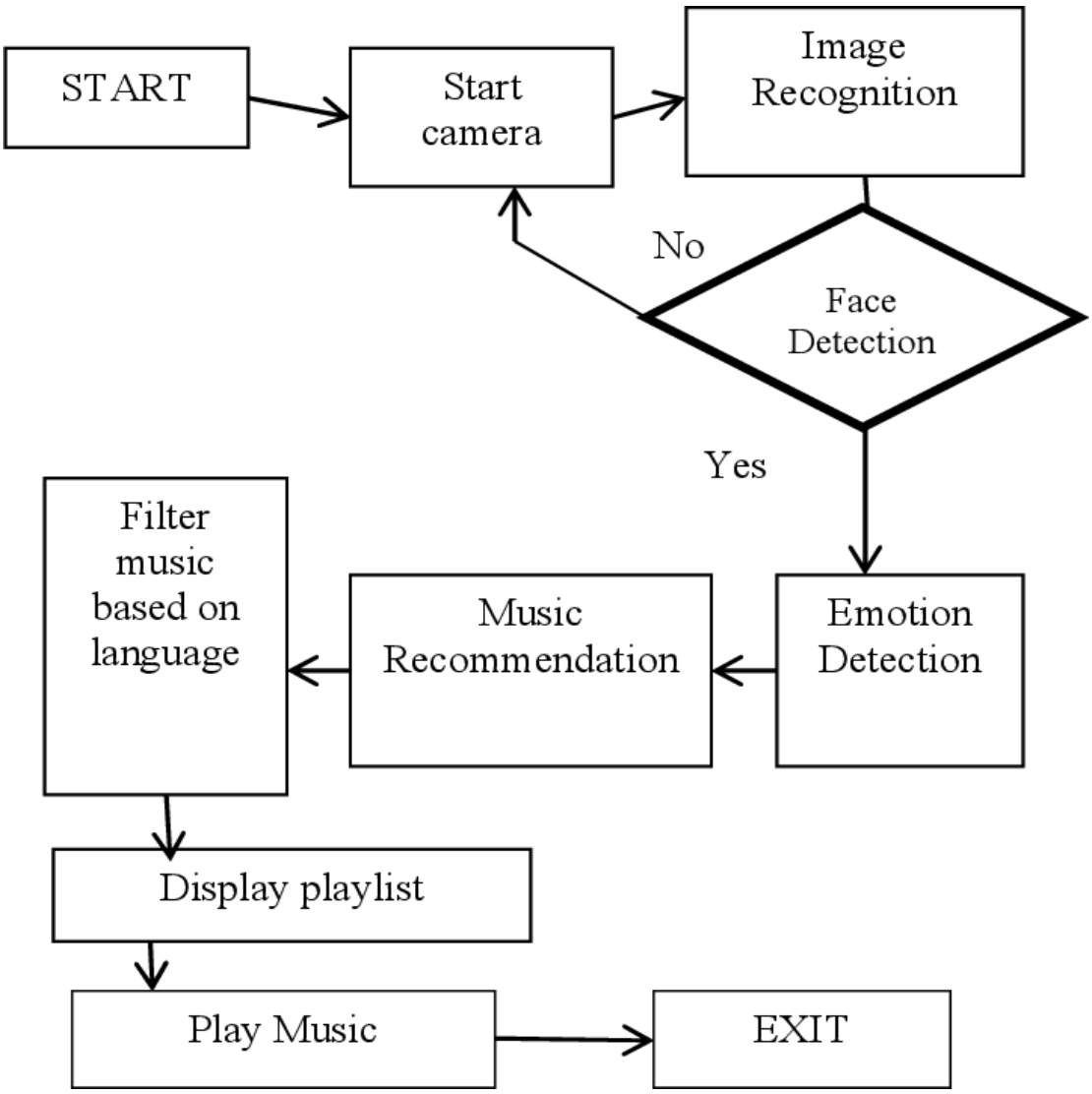
The proposed system follows a modular architecture that combines computer vision, deep learning, and multimedia processing. The design contains the following essential components:

3.3.1 Key Components

1. **Input Layer** – Captures real-time video feed from the webcam.
2. **Preprocessing Module** – Converts input to grayscale, resizes frames, and removes noise.
3. **Face Detection Module** – Identifies and crops the user's face region.
4. **Emotion Recognition Model** – Uses CNN/ResNet to classify the facial expression.
5. **Emotion Processing Layer** – Converts predicted class into an emotion label.
6. **Audio Curation Engine** – Selects a playlist matching the emotion.
7. **User Interface Module** – Displays detected emotion and curated audio suggestions.

This design ensures real-time performance, modularity, and ease of deployment across desktop and mobile platforms.

3.3 UML Diagram



CHAPTER 4
RESULTS AND DISCUSSION

CHAPTER 4

RESULTS AND DISCUSSION

Fig 4.1 Capturing Face

Fig 4.2 Result

CHAPTER 5
CONCLUSION

CHAPTER 5

CONCLUSION

The proposed AI-Driven Facial Expressive State Audio Curation System successfully demonstrates how computer vision and deep learning can be combined to create an intelligent, emotion-aware multimedia experience. By integrating facial expression recognition with an automated audio recommendation module, the system effectively identifies real-time emotional states and curates suitable music without requiring user input. The use of CNN/ResNet-based models ensures reliable emotion detection, while the emotion-to-audio mapping engine delivers personalized and mood-adaptive music suggestions.

Overall, the system achieves its objective of providing fast, accurate, and user-friendly emotion-based audio curation. It shows strong potential for use in entertainment applications, mental-wellness platforms, and personalized media systems. The work establishes a foundation for affective computing solutions that enhance user engagement through intelligent, adaptive interaction.

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